Customer Behaviour Analysis

A structured retail/e-commerce dataset used to understand purchasing patterns, optimize marketing strategies, and increase customer lifetime value through Python EDA, SQL analysis, and Power BI visualization.

Made with GAMMA

About the Dataset

Type

Retail / E-commerce · Structured Tabular Data at Customer Transaction Level

Numerical Features

Age, Purchase Amount, Previous Purchases

Categorical Features

Category, Payment Method, Season

Purpose

Customer behavior analysis + business decision-making

Made with GAMMA

Why This Data Matters

In e-commerce, companies track every interaction — what customers buy, how often, how much they spend, and whether they respond to discounts. This data helps businesses:

High-Value Customers

Identify top spenders

Purchase Patterns

Understand buying behavior

Discount Strategy

Optimize promotions

Marketing Campaigns

Improve targeting

Customer Lifetime Value

Increase CLV

Made with GAMMA

Business Questions

How can a retail company understand customer purchasing behavior to increase revenue, improve retention, and optimize marketing strategies?

- → Which category generates the highest revenue?
- → Are discounts increasing purchase value?
- → Revenue contribution by gender & age group?
- → Do subscribed customers spend more?
- → Top/bottom products by review ratings?
- → Identify high-spending discount users
- → Avg purchase: Standard vs Express shipping
- → Products with highest discount usage
- → Customer segmentation analysis
- → Repeat buyers & subscription relationship

Made with GAMMA

EDA: Data Loading & Exploration

Data Loading

Imported **pandas** and loaded the dataset using `pd.read_csv('customer_shopping_behavior.csv')`. Key columns: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount, Review Rating, Subscription Status, Discount Applied, Previous Purchases, and more.

Exploration Commands

- `df.head()` / `df.tail()`
- `df.describe(include='all')`
- `df.info()` / `df.dtypes`
- `df.isnull().sum()`
- `df.duplicated().sum()`

Made with GAMMA

Data Cleaning & Preprocessing

1

Category Correction

Created a mapping dictionary to align **Item Purchased** with correct categories (Clothing, Accessories, Electronics, Footwear, Outerwear).

2

Missing Values

Size: category-based fill (e.g., "Not Applicable" for Electronics). Review Rating & Purchase Amount: group-by mean. Previous Purchases: filled with 0.

3

Duplicates & Standardization

Dropped duplicate Customer IDs. Standardized column names: lowercase with underscores.

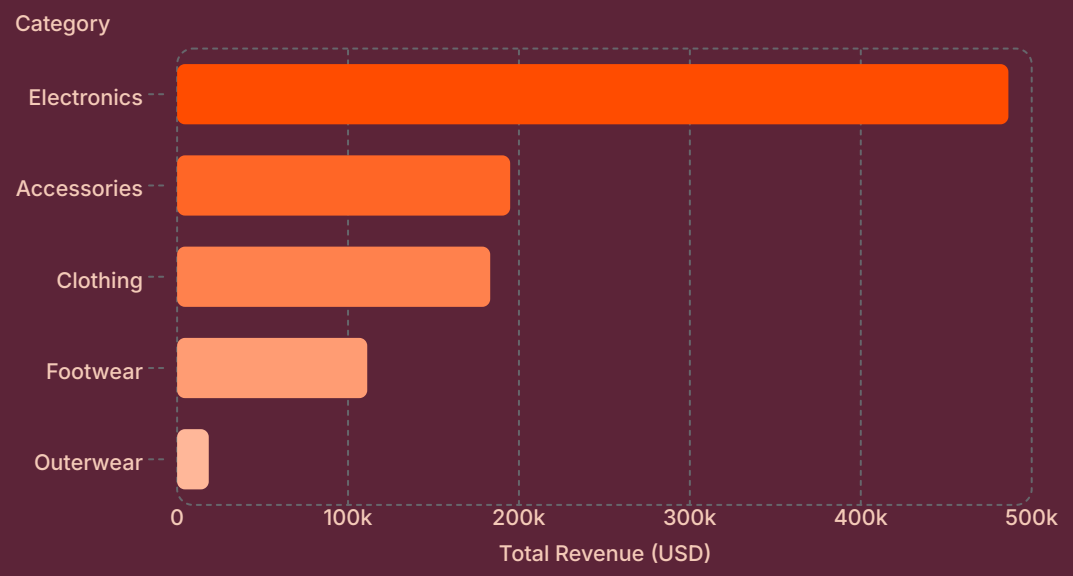
4

Export to SQL Server

Used **SQLAlchemy** + **pyodbc** to upload cleaned data to SQL Server table `customer_behaviour_analysis`.

Made with GAMMA

SQL Analysis: Revenue Insights



Key Findings

- **Electronics** leads revenue at $486K — nearly 2.5× Accessories
- Discounts **increase avg purchase** from $182 → $219, but total revenue is slightly lower
- Male customers: $391K · Female: $352K · Other: $251K
- Age 51+ contributes the most: **$399K (40.2%)**, followed by 35–50 at $267K

Made with GAMMA

SQL Analysis: Customers & Products

3085

Loyal Customers

Segmented by previous purchases

561

New Customers

First-time or low-frequency buyers

1354

Returning Customers

Moderate purchase history

270.58

Avg Spend (Subscribers)

vs $165.72 for non-subscribers

Top Products by Discount Usage

Laptop (51.7%), Phone (51.5%), Headphones (51.2%) — all over 50% discount rate.

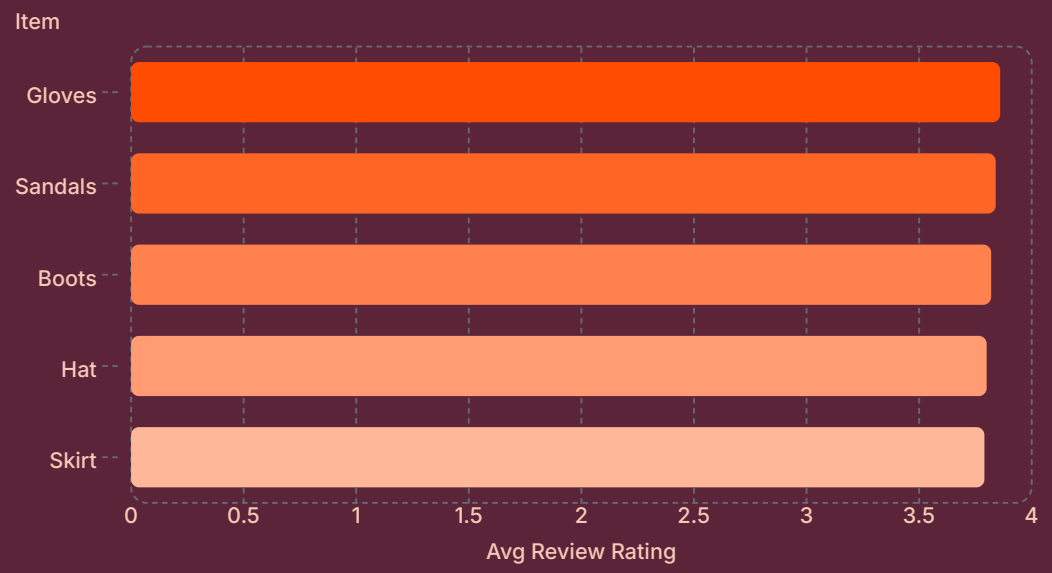
Repeat Buyers & Subscriptions

Repeat buyers with subscriptions: **1,190**. Strong link between repeat purchases and subscription status.

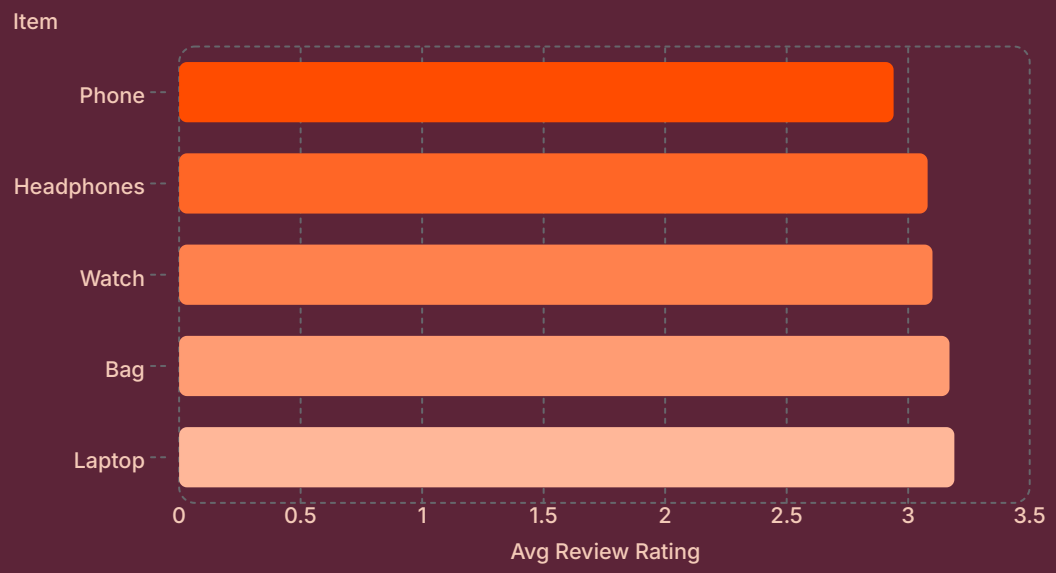
Made with GAMMA

SQL Analysis: Ratings & Shipping

Top 5 Products by Rating



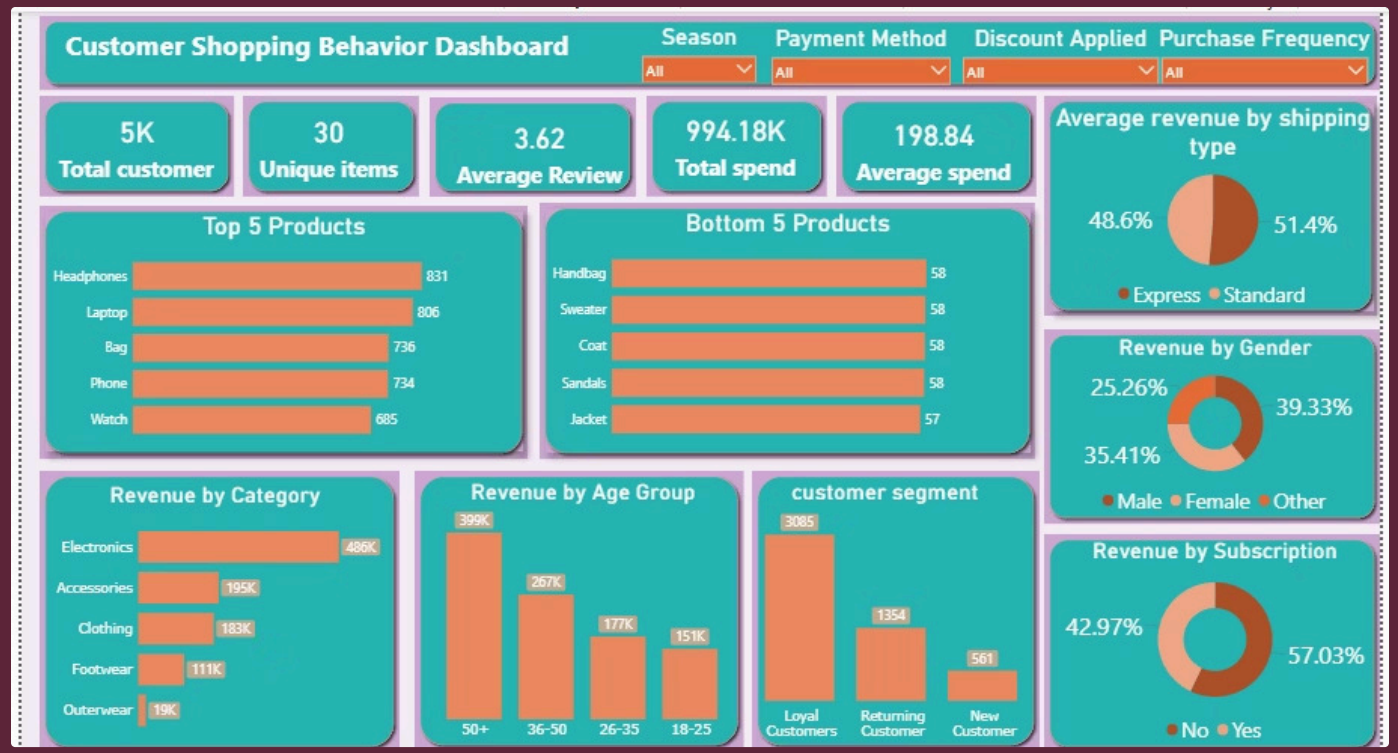
Bottom 5 Products by Rating



Express shipping has a slightly higher avg purchase ($358.12) vs Standard ($340.57), with nearly equal order volumes (~1,200 each).

Made with GAMMA

Power BI Dashboard



Key KPIs

Top Revenue by Category

Electronics: $486K · Accessories: $195K · Clothing: $183K · Footwear: $111K · Outerwear: $19K

Top Items by Revenue

Headphones: $138K · Phone: $126K · Bag: $121K · Laptop: $115K · Watch: $108K

5,000
Total Customers

30
Unique Items

3.62
Avg Rating

$994K
Total Spend

Made with GAMMA